UNIVERSITY OF ENGINEERING AND TECHNOLOGY (UET), LAHORE

SEMESTER PROJECT

PROJECT NAME:

PROJECT MANAGEMENT SYSTEM

(Digital Project Manager)

Submitted by:

NAME:

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ROLL NO:

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Program:

BS Computer Science (Section-B)

Course:

Programming Fundamentals

DETAILS

Introduction:

Introduction to Project Management System

- This project is a console-based Project Management System developed in C++.
- It is designed to help manage tasks, projects, and employees efficiently within an organization.
- The system provides two types of users:
    - **Manager**
    - **Employee**

Objective:

The objective of this project is to help different developers working all around the world to manage their projects properly. They can keep track of which tasks are assigned to them and also deadlines of those tasks. It is also very helpful for managers as they can assign project to different employees and those employees can work according to the conditions described.

Project Functional Requirements

Users:

- Manager
- Employee

Manager Options:

To access manager options you first need to enter username and password.

- Manager can create project entering details like Title, Task, Deadline, Priority, Status.
- Manager can update task of any project.
- Manager can update any project.
- Manager can sort project according to deadline.
- Manager can view list of all projects.
- Manager can delete any project.
- Manager can assign project to any employee.
- Manager can add employees.
- Manager can view list of all employees.

Employees:

- Employees can select who they are.
- Employees can view project assigned to them.
- Employees can update status of project.
- Employees can mark project as complete.
- Employees can view deadlines of their projects.
- Employees can search an project by a task.
- Employees can check number of projects assigned to them.

WIRE-FRAMES

- Main-Screen:

```
---------------------------------------------
---------------------------------------------
-------------PROJECT MANAGER-----------------
---------------------------------------------
---------------------------------------------



----------------USERS---------------------

1.MANAGER
2.EMPLOYEE
3.EXIT!
Select an option: |
```

- Manager Menu:

```
--------MANAGER-MENU---------
1.Create Project
2.Update Task
3.Update Project
4.Sort by Deadline
5.View list
6.Delete project
7.Assign Project
8.Add Employee
9.View Employee list
10.Exit
Select an option (1 - 9): |
```

- Employees Menu:

```
----------EMPLOYEE MENU----------
EMPLOYEES LIST
1 Muhammad
2 Shoaib
Choose Who you are: (1-2): |
```

- Employees Options:

```
WELCOME Muhammad!!!

Choose what to do:
1.View Assigned project
2.Update Status
3.Mark Complete
4.View Deadlines
5.Search task
6.Show Number of projects
7.Exit
Select option (1-7): |
```

DATA STRUCTURES

We are storing data using parallel arrays.

Following are the data structures used in this project:

string tasks[totalTasks] (Stores tasks assigned to projects)

string titleA[totalTasks] (Stores titles of projects)

int deadlineA[totalTasks] (Stores deadline of each project)

string priorityA[totalTasks] (Stores priority of each project)

string statusA[totalTasks] (Stores Status of each project)

string employees[10] (Stores names of employees working)

SOURCE CODE

```
#include <iostream>
#include <conio.h>
using namespace std;
main()
{

    int totalTasks = 100;
    int count = 5;
    int employeeCount = 2;

    string tasks[totalTasks] = {" ", "EditVideo", "LogoDesign", "SetupPC_OS",
"Testing_OS"};
    string titleA[totalTasks] = {" ", "Market", "Brand", "IT", "QA"};
    int deadlineA[totalTasks] = {0, 12, 3, 20, 2};
    string priorityA[totalTasks] = {" ", "Medium", "High", "Low", "null"};
    string statusA[totalTasks] = {" ", "InProgress", "Pending", "Done", "Pending"};
    string employees[10] = {" ", "Muhammad", "Shoaib", "Saleem"};

    int assignedTo[totalTasks] = {0, 1, 2, 1, 2};

    while (true)
    {
        system("cls");
        cout << "----------------------------------------------\n";
        cout << "----------------------------------------------\n";
        cout << "--------------PROJECT MANAGER-----------------\n";
        cout << "----------------------------------------------\n";
        cout << "----------------------------------------------\n";

        cout << endl
             << endl
             << endl;
        cout << "-----------------USERS--------------------\n";
        cout << endl;

        cout << "1.MANAGER\n";
        cout << "2.EMPLOYEE\n";
```

```
cout << "3.EXIT!\n";
cout << "Select an option: ";
int userOption;
cin >> userOption;
system("cls");
if (userOption == 1)
{
    // Manager Options
    for (int i = 0; i < 4; i++)
    {
        if (i < 3)
        {
            cout << "Enter Credentials: \n";
            cout << "Enter Username: ";
            string userName;
            cin >> userName;

            cout << "Enter Password: ";
            string pass;
            cin >> pass;
            system("cls");
            if (userName == "admin" && pass == "123")
            {
                int managerOption;

                while (true)
                {
                    system("cls");
                    cout << "--------MANAGER-MENU---------\n";
                    cout << "1.Create Project\n";
                    cout << "2.Update Task\n";
                    cout << "3.Update Project\n";
                    cout << "4.Sort by Deadline\n";
                    cout << "5.View list\n";
                    cout << "6.Delete project\n";
                    cout << "7.Assign Project\n";
                    cout << "8.Add Employee\n";
                    cout << "9.View Employee list\n";
                    cout << "10.Exit\n";
                    cout << "Select an option (1 - 9): ";

                    cin >> managerOption;
```

```
if (managerOption == 1)
{
    // Create Project Code
    system("cls");

    cout << "Enter details about Project: \n";
    cout << "Enter Title: ";
    string title;
    cin >> title;

    cout << "Enter Task: ";
    string task;
    cin >> task;

    cout << "Enter Deadline: ";
    int deadline;
    cin >> deadline;

    cout << "Enter Priority: ";
    string priority;
    cin >> priority;

    cout << "Enter Status: ";
    string status;
    cin >> status;
    system("cls");
    cout << "ALL DONE!\n";
    cout << "Press any key to continue! ";
    getch();

    tasks[count] = task;
    titleA[count] = title;
    deadlineA[count] = deadline;
    priorityA[count] = priority;
    statusA[count] = status;

    count++;
}
else if (managerOption == 2)
{
    // update Task Code
    system("cls");
```

```
                        cout << "Sr\tName\tDeadline(in
weeks)\tTask\t\tPriority\tStatus\n";
                        for (int i = 1; i < count; i++)
                        {
                            if (deadlineA[i] != 0)
                            {
                                cout << i << "\t" << titleA[i] << "\t\t" << deadlineA[i]
<< "\t\t" << tasks[i] << "\t" << priorityA[i] << "\t\t" << statusA[i] << endl;
                            }
                        }
                        int select;
                        cout << "Select Project: ";
                        cin >> select;

                        cout << "Enter Task: ";
                        string task;
                        cin.ignore();
                        getline(cin, task);

                        tasks[select] = task;
                    }
                    else if (managerOption == 3)
                    {
                        // Update Project
                        system("cls");
                        cout << "Select Project Update: \n";

                        cout << "Sr\tName\tDeadline(in
weeks)\tTask\t\tPriority\tStatus\n";
                        for (int i = 1; i < count; i++)
                        {
                            if (deadlineA[i] != 0)
                            {
                                cout << i << "\t" << titleA[i] << "\t\t" << deadlineA[i]
<< "\t\t" << tasks[i] << "\t" << priorityA[i] << "\t\t" << statusA[i] << endl;
                            }
                        }
                        int select;
                        cout << "Select Project: ";
                        cin >> select;

                        cout << "\n";
```

```
    cout << "Enter details about Project to Update: \n";
    cout << "Enter Title: ";
    string title;
    cin >> title;

    cout << "Enter Task: ";
    string task;
    cin >> task;

    cout << "Enter Deadline: ";
    int deadline;
    cin >> deadline;

    cout << "Enter Priority: ";
    string priority;
    cin >> priority;

    cout << "Enter Status: ";
    string status;
    cin >> status;
    system("cls");
    cout << "Project Updated Successfully!\n";
    cout << "Press any key to continue! ";
    getch();

    tasks[select] = task;
    titleA[select] = title;
    deadlineA[select] = deadline;
    priorityA[select] = priority;
    statusA[select] = status;
}
else if (managerOption == 4)
{
    // Sort by Deadline Code
    for (i = 1; i < count; i++)
    {
        for (int j = i + 1; j < count; j++)
        {
            if (deadlineA[i] > deadlineA[j])
            {
                int deadtemp = deadlineA[i];
                deadlineA[i] = deadlineA[j];
                deadlineA[j] = deadtemp;
```

```
                                string taskTemp = tasks[i];
                                tasks[i] = tasks[j];
                                tasks[j] = taskTemp;

                                string titleTemp = titleA[i];
                                titleA[i] = titleA[j];
                                titleA[j] = titleTemp;

                                string priorityTemp = priorityA[i];
                                priorityA[i] = priorityA[j];
                                priorityA[j] = priorityTemp;

                                string statustemp = statusA[i];
                                statusA[i] = statusA[j];
                                statusA[j] = statustemp;
                            }
                        }
                    }
                    cout << "Projects in ascending order: \n";

                    cout << "Sr\tName\tDeadline(in
weeks)\tTask\t\tPriority\tStatus\n";
                    for (int i = 1; i < count; i++)
                    {
                        if (deadlineA[i] != 0)
                        {
                            cout << i << "\t" << titleA[i] << "\t\t" << deadlineA[i]
<< "\t\t" << tasks[i] << "\t" << priorityA[i] << "\t\t" << statusA[i] << endl;
                        }
                    }
                    cout << "press any key to continue! ";
                    getch();
                }

                else if (managerOption == 5)
                {
                    // View Project list
                    system("cls");
                    cout << "The list of Projects is: \n\n";

                    cout << "Sr\tName\tDeadline(in
weeks)\tTask\t\tPriority\tStatus\n";
```

```
                    for (int i = 1; i < count; i++)
                    {
                        if (deadlineA[i] != 0)
                        {
                            cout << i << "\t" << titleA[i] << "\t\t" << deadlineA[i]
<< "\t\t" << tasks[i] << "\t" << priorityA[i] << "\t\t" << statusA[i] << endl;
                        }
                    }

                    cout << "Press any to continue ";
                    getch();
                }
                else if (managerOption == 6)
                {
                    // Delete Task Code
                    system("cls");
                    cout << "Sr\tName\tDeadline(in
weeks)\tTask\t\tPriority\tStatus\n";
                    for (int i = 1; i < count; i++)
                    {
                        if (deadlineA[i] != 0)
                        {
                            cout << i << "\t" << titleA[i] << "\t\t" << deadlineA[i]
<< "\t\t" << tasks[i] << "\t" << priorityA[i] << "\t\t" << statusA[i] << endl;
                        }
                    }
                    int select;
                    cout << "Select Project to delete: ";
                    cin >> select;

                    tasks[select] = " ";
                    titleA[select] = " ";
                    deadlineA[select] = 0;
                    priorityA[select] = " ";
                    statusA[select] = " ";

                    for (i = select; i < count; i++)
                    {
                        tasks[i] = tasks[i + 1];
                        titleA[i] = titleA[i + 1];
                        deadlineA[i] = deadlineA[i + 1];
                        priorityA[i] = priorityA[i + 1];
                        statusA[i] = statusA[i + 1];
```

```
                    }
                    cout << "Task Deleted Successfully!\n";
                    cout << "Press any key to continue!";
                    getch();
                }
                else if (managerOption == 7)
                {
                    // Assign Project Code
                    system("cls");
                    cout << "Select Project You Wanna Assign: \n\n";
                    cout << "Sr\tName\tDeadline(in
weeks)\tTask\t\tPriority\tStatus\n";
                    for (int i = 1; i < count; i++)
                    {
                        if (deadlineA[i] != 0)
                        {
                            cout << i << "\t" << titleA[i] << "\t\t" << deadlineA[i]
<< "\t\t" << tasks[i] << "\t" << priorityA[i] << "\t\t" << statusA[i] << endl;
                        }
                    }
                    cout << "Select Project you wanna assign: ";

                    int select;
                    cin >> select;
                    system("cls");

                    cout << "Select employee you wanna assign project to: \n";

                    for (int i = 1; i <= employeeCount; i++)
                    {
                        cout << i << " " << employees[i] << endl;
                    }
                    cout << "Choose Employee: (1-" << employeeCount << ")";
                    int employeeSelect;
                    cin >> employeeSelect;

                    assignedTo[select] = employeeSelect;

                    cout << "Project Successfully Assigned!";
                }
                else if (managerOption == 8)
                {
                    // Add Employee Code
```

```
                system("cls");
                cout << "Enter Details About Employee: \n";

                cout << "Enter Employee Name: ";
                string employee;
                cin >> employee;
                system("cls");

                employeeCount++;
                employees[employeeCount] = employee;

                cout << "Employee Added Successfully!";
            }
            else if (managerOption == 9)
            { // Employees list code
                system("cls");
                cout << "EMPLOYEES LIST\n";
                for (int i = 1; i <= employeeCount; i++)
                {
                    cout << i << " " << employees[i] << endl;
                }

                cout << "Press any key to continue! ";
                getch();
            }

            else if (managerOption == 10)
            {
                break;
            }
        }
        break;
    }
    else
    {
        cout << "Username or password is wrong!\n";
    }
}
else
{
    cout << "TOO MANY ATTEMPTS! \n";
    cout << "Press any key to continue!";
    getch();
```

```
        }
    }
}

else if (userOption == 2)
{ // Employee Menu
    cout << "----------EMPLOYEE MENU----------\n";
    cout << "EMPLOYEES LIST\n";
    for (int i = 1; i <= employeeCount; i++)
    {
        cout << i << " " << employees[i] << endl;
    }
    cout << "Choose Who you are: (1-" << employeeCount << "): ";
    int employeeSelect;
    cin >> employeeSelect;
    while (true)
    {

        system("cls");
        cout << "WELCOME " << employees[employeeSelect] << "!!!" << endl
             << endl;
        // Worker Options
        cout << "Choose what to do: \n";
        cout << "1.View Assigned project\n";
        cout << "2.Update Status\n";
        cout << "3.Mark Complete\n";
        cout << "4.View Deadlines\n";
        cout << "5.Search task\n";
        cout << "6.Show Number of projects\n";
        cout << "7.Exit\n";

        cout << "Select option (1-7): ";

        int employeOption;
        cin >> employeOption;

        if (employeOption == 1)
        { // Assigned Tasks
            cout << "Your Assigned Tasks:\n";
            cout << "Sr\tName\tDeadline(in weeks)\tTask\t\tPriority\tStatus\n";
            for (int i = 1; i < count; i++)
            {
                if (assignedTo[i] == employeeSelect)
```

```
                    {
                        if (deadlineA[i] != 0)
                        {
                            cout << i << "\t" << titleA[i] << "\t\t" << deadlineA[i] <<
"\t\t" << tasks[i] << "\t" << priorityA[i] << "\t\t" << statusA[i] << endl;
                        }
                    }
                }
                cout << "Press any key to continue!";
                getch();
            }
            else if (employeOption == 2)
            { // Task Status Update
                cout << "Your Projects are\n";
                cout << "Sr\tName\tDeadline(in weeks)\tTask\t\tPriority\tStatus\n";
                for (int i = 1; i < count; i++)
                {
                    if (assignedTo[i] == employeeSelect)
                    {
                        if (deadlineA[i] != 0)
                        {
                            cout << i << "\t" << titleA[i] << "\t\t" << deadlineA[i] <<
"\t\t" << tasks[i] << "\t" << priorityA[i] << "\t\t" << statusA[i] << endl;
                        }
                    }
                }

                cout << "Select Project you wanna update Status of: ";

                int select;
                cin >> select;

                cout << "Enter Status: ";
                string status;
                cin.ignore();
                getline(cin, status);
                statusA[select] = status;
            }
            else if (employeOption == 3)
            { // Marks Project as complete
                cout << "Your Projects are\n";
                cout << "Sr\tName\tDeadline(in weeks)\tTask\t\tPriority\tStatus\n";
                for (int i = 1; i < count; i++)
```

```
                {
                    if (assignedTo[i] == employeeSelect)
                    {
                        if (deadlineA[i] != 0)
                        {
                            cout << i << "\t" << titleA[i] << "\t\t" << deadlineA[i] <<
"\t\t" << tasks[i] << "\t" << priorityA[i] << "\t\t" << statusA[i] << endl;
                        }
                    }
                }

                cout << "Select Project you mark as complete: ";

                int select;
                cin >> select;

                statusA[select] = {"Complete"};

                cout << "Project is Marked as complete\n";
                cout << "Press any key to continue";
                getch();
            }
            else if (employeOption == 4)
            { // Show deadlines
                cout << "Deadlines of your projects are: \n";
                cout << "Sr\tDeadline(in weeks)\n";
                for (int i = 1; i < count; i++)
                {
                    if (assignedTo[i] == employeeSelect)
                    {
                        if (deadlineA[i] != 0)
                        {
                            cout << i << "\t" << deadlineA[i] << endl;
                        }
                    }
                }
                cout << "press any key to continue!";
                getch();
            }
            else if (employeOption == 5)
            { // Search by task
                cout << "Enter Task you wanna search: ";
                string task;
```

```
                cin.ignore();
                getline(cin, task);

                for (int i = 1; i <= count; i++)
                {
                    if (assignedTo[i] == employeeSelect)
                    {
                        if (tasks[i] == task)
                        {
                            if (deadlineA[i] != 0)
                            {
                                cout << "Sr\tName\tDeadline(in
weeks)\tTask\t\tPriority\tStatus\n";
                                cout << i << "\t" << titleA[i] << "\t\t" << deadlineA[i] <<
"\t\t" << tasks[i] << "\t" << priorityA[i] << "\t\t" << statusA[i] << endl;
                                break;
                            }
                        }
                    }
                    else if (i == count)
                    {
                        cout << "TASK NOT FOUND!\n";
                    }
                }

                cout << "Press any key to continue!";
                getch();
            }
            else if (employeOption == 6)
            { // Number of tasks
                int tasksNumber;
                for(int i = 0; i<count; i++){
                    if(assignedTo[i] == employeeSelect){
                        tasksNumber++;
                    }
                }
                cout<<"The number of tasks assigned to you are:
"<<tasksNumber<<endl;
                cout<<"Press any key to continue!";
                getch();
                tasksNumber = 0;

            }else if (employeOption == 7)
```

```
            {
                break;
            }
        }
    }
    else if (userOption == 3)
    {
        // Program Ends
        cout << "Program Ended\n";
        cout << "Thanks for using this Program.\n";
        cout << "Press any key to continue!";
        getch();
        break;
    }
    else
    {
        // Invalid Option
        cout << "Invalid Option! Please select correct option (1-3).\n";
        cout << "press any key to continue!";
        getch();
    }
  }
}
```